MIL-F-45090B (ORD)

3 April 1961

SUPERSEDING
MIL-F-45090A (ORD)
20 November 1959

MILITARY SPECIFICATION

FUZE, POINT DETONATING, M505A1

PARTS FOR

1. SCOPE

1.1 This specification covers parts for the M505A1 fuze for use in 20mm ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition

MIL-P-10025 - Packaging and Marking for Domestic Shipment of Inert Ammunition Components; General Specification For.

STANDARDS

FEDERAL

Federal Test Method Standard No. 151 - Metals; Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 - Inspection Terms and Definitions

FSC 1305

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DRAWINGS

ORDNANCE CORPS

D7258818 - Fuze, PD, M505A1 Assembly
GL 7259308 - Index to List of Inspection Gages and Parts Gaged On: Fuze, PD, M505A1, Metal Parts
D7259320 - Spring Checker Assembly

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Sampling Tables and Procedures for Continuous Sampling by Attributes

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on applicable detail drawings referenced on Drawing D7258818, and with all requirements shown in applicable specifications.

3.3 Corrosion resistance.-The body, booster holder, cover, and safety spring retainer shall show no evidence of white salts or red rust when tested as specified in 4.4.1. Exposure time shall be as shown on the applicable drawings.

3.4 Spring displacement.-The body, spring and spring retainer assembly shall permit a 0.723 inch penetration of a test ram positioned against the safety spring when loaded with not less than 4.3 pounds nor more than 13.0 pounds. After release of the load, the safety spring shall not exceed maximum locational limit specified on the applicable component assembly drawing.

3.5 Preproduction sample.-A preproduction sample, as specified in 4.2, shall be required prior to the start of regular production.

3.6 Workmanship.-All parts shall be free from chips, dirt, grease, corrosion, and other foreign material. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents used.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions

4.1.1 Contractor inspection.-Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. Reference shall be made to MIL-STD-109 for definition of inspection terms used herein.

4.1.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor shall not be restricted to a particular quality assurance system. However, the system adopted must be one which consistently results in material meeting the requirements as evidenced by the evaluation specified in 4.1.2.1. In all testing inspection, the use of the equipment specified and the methods of tests prescribed in this specification and other documents referenced herein shall be mandatory. Results of all examinations and tests performed under this quality assurance system shall be made available to the Government. The contractor shall notify the Government of and obtain approval for any change to the written procedure that affects the degree of assurance required by this specification and other documents referenced herein.

4.1.1.2 Submission of data.-At the time the product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with quality assurance provisions and requirements as required by the contractual documents.

b. Certificates of analysis including results of tests on materials used in the lot.

4.1.1.3 The certificate shall be signed by a responsible agent of the certifying organization and shall be substantiated by evidence of the agent's authority to bind his principal.

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4.1.2 Government verification

4.1.2.1 At the beginning of the contract and until such time as the contractor has operated under an established and approved quality assurance system which has resulted in an acceptable product, Government verification will consist of (1) performance of all inspections and tests listed in the detail specification and other documents referenced therein, which are not reserved for sole performance by the contractor, (2) monitoring of those inspections reserved for sole performance by the contractor, and (3) monitoring of the contractor's quality assurance operations. At no time will complete Government verification be recuced to less than that permitted by MIL-STD-105 or ORD-M608-11 except under those conditions prescribed in 4.1.2.2.

4.1.2.2. Under the following conditions further reduced Government verification may be initiated on a complete or individual operation basis.

a. On a voluntary basis the contractor includes in his quality assurance provisions written quality control procedures evaluated and approved by the Government.

b. Verification inspection by the Government in accordance with 4.1.2.1 indicates that the quality of product continues to be equal to or better than that required by this specification and the referenced documents.

c. The contractor's and the Government's inspection results are found to be consistent with each other within the limits of sampling error.

4.1.2.3 At such time as the contractor meets the conditions of 4.1.2.2 above, Government verification will consist of, as a minimum, inspection for critical defects, monitoring of contractor's quality assurance system for strict compliance to the written plan and review of contractor records on all inspections performed. If the contractor's quality deteriorates or a departure from the written plan is noted, the Government inspector will immediately notify the contractor to take corrective action and until the deficiencies have been corrected, Government verification in accordance with 4.1.2.1 will be performed on those characteristics involved to determine acceptability of the product. If corrective action is not taken by the contractor, as requested by the Government, acceptance of the product may be suspended. Upon correction of the deficiencies, the Government will continue inspection as prescribed in 4.1.2.1 until contractor has requalified for reduced Government verification as prescribed in 4.1.2.2.

4.1.2.4 Critical defects.-Government inspection for critical defects will be conducted at all times in accordance with MIL-G-2550.

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4.2 Preproduction sample.- Prior to the start of regular production, the contractor shall submit preproduction sample(s) manufactured by the same methods and equipment that will be utilized in performing the work under the contract. The contracting officer will furnish instructions for shipment of the sample(s) to a designated Government laboratory for examination and test.

4.2.1 Seventy-five (75) sets of component parts and twenty-five (25) body, spring and spring retainer assemblies will be sent to and examined by a Government laboratory for compliance with the requirements of the applicable drawings and this specification.

4.2.2 Results of the examination and test will be furnished to the contractor. Approval or disapproval of the preproduction sample(s) will be given by the responsible procuring activity. Instructions for the submission of new sample(s) in case of disapproval of the original sample will be issued by the procuring activity. Regular production shall not proceed until the preproduction sample(s) has been approved.

4.3 Acceptance Inspection

4.3.1 Lot formation.- A lot shall consist of fuze metal parts and assemblies produced by one manufacturer in one unchanged process in accordance with the same drawings and drawing revisions, same specification and specification revision,under conditions of continuous production free from interruptions other than those due to the end of a shift, day, or work week. (See 6.2).

4.3.2 Examination.- Inspection will be performed as specified in specification MIL-G-2550 and as indicated below. Lot sampling and acceptance criteria will be in accordance with Standard MIL-STD-105 or Publication ORD-M608-11, as appropriate.

4.3.2.1 Judgement for lot acceptance shall be based on either the individual defect characteristic or on classes of defect characteristics with individual defect characteristic limitations, as approved by the procuring activity prior to the initiation of production, with the following AQL's assigned (see 6.3):

4.3.2.1.1. Individual defect characteristic basis

Major Defects	0.25%
Minor Defects	0.40%

4.3.2.1.2 Defect characteristic class basis

Major Defects	1.50%
Minor Defects	2.50%

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Individual defect characteristic AQL's associated with the class basis, shall be

Major Defects	0.40%
Minor Defects	0.65%

4.3.2.2 The contractor shall examine the parts for the critical defects listed in 4.3.4 and shall remove those parts found to be defective.

4.3.3 Test inspection.-The tests indicated below shall be performed in accordance with the methods specified in 4.4. The number of test samples for each lot shall be as shown in Table I. Acceptance criteria shall be as specified.

TABLE I

Test	Part	Number of samples
Corrosion resistance	Body	25
	Booster holder	25
	Cover	25
	Safety spring retainer	25
Spring displacement	Body, spring and spring retainer assembly	MIL-STD-105
Hardness	Cover	25

4.3.3.1 Corrosion resistance.-Twenty-five of the prescribed parts shall be selected from each lot and tested as specified in 4.4.1. If none of the parts being tested fail to meet the requirements for corrosion resistance the parts are acceptable. If any part of the sample tested fails to meet the requirements for corrosion resistance, the lot shall be rejected except that in the event of one or two failures to meet the requirements for corrosion resistance, a retest of fifty parts is authorized. The parts shall be finally rejected if in the combined test and retest a total of three or more parts fail to meet the corrosion resistance requirements.

4.3.3.2 Spring displacement.-Any body assembled with safety spring and safety spring retainer, which fails to meet the load and spring tip location requirements of 3.4 shall be removed from the lot.

4.3.3.3 Spring embrittlement.-Twenty-five (25) springs shall be selected from each lot and tested as specified in 4.4.3. If any spring cracks or fractures, the parts shall be finally rejected.

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4.3.3.4 Hardness.-Twenty-five of the fuze covers shall be selected from each lot and tested as specified in 4.4.4. If none of the parts being tested fail to meet the requirements for hardness the parts are acceptable. If any part of the sample tested fails to meet the requirements for hardness, the lot shall be rejected except that in the event of one or two failures to meet the requirements for hardness, a retest of fifty fuze covers is authorized. The covers shall be finally rejected if in the combined test and retest a total of three or more covers fail to meet the hardness requirements.

4.3.4 Classification of defects

4.3.4.1 Booster holder (see Dwg. C725886 covering a detail of Dwg. D7258818)

	Categories and defects	Method of inspection
Major		
1.	Length from forward edge of flange to top of .37700 ball	Gage
2.	Crimping groove missing	Visual
3.	Major diameter of thread	Gage
4.	Pitch diameter of thread	Gage
5.	Concentricity of outside diameter of body with pitch diameter of thread	Gage
6.	Concentricity of flange diameter with pitch diameter of thread	Gage
7.	Concentricity of .37700 ball with pitch diameter of thread	Gage
8.	Body diameter forward of thread, max	Gage
9.	Length from forward edge of flange to end of thread, max	Gage
10.	Length of thread, min.	Gage
11.	Finish of spherical radius	Measuring device
12.	Foreign matter	Visual (see 4.3.5.2)
13.	Damage	Visual
Minor		
41.	Diameter of flash hole	Gage
42.	Diameter of booster cavity, min.	Gage
43.	Depth of booster cavity from forward edge of flange	Gage
44.	Diameter of flange, max	Gage
45.	Protective coating damaged or missing	Visual
46.	Burr	Visual
47.	Finish except spherical radius	Visual

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Minor (cont)

48. Diameter of undercut, rear of thread----------Gage
49. Height, crimping lip, min----------Gage

4.3.4.2 Rotor (see Dwg. C7258827, covering a detail of Dwg.D7258818)

Categories and defects	Method of inspection
Major	
1. Outside diameter	Gage
2. Crimping groove missing	Visual
3. Depth of crimping groove, min.	Gage
4. Burr above spherical surface	Visual
5. Finish of spherical surface	Measuring device
6. Length from crimping lip to apex of reference cone, max.	Gage
7. Foreign matter	Visual (see 4.3.5.2)
Minor	
41. Diameter of flash hole, min.	Gage
42. Diameter of detonator cavity, min.	Gage
43. Length from bottom of detonator cavity to apex of reference cone	Gage
44. Finish except spherical surface	Visual
45. Flat on flash hole	Visual-Gage
46. Burr except above spherical surface	Visual
47. Concentricity of detonator cavity with outside diameter	Gage
48. Concentricity of flash hole with detonator cavity	Gage
49. Length from crimping lip to apex of reference cone, min.	Gage

4.3.4.3 Spring, rotor safety (see Dwg D7258824, covering a detail of Dwg. D7258818)

Categories and defects	Method of inspection
Major	
1. Profile of spring arm from base to tip	Gage
2. Protective coating damaged or missing	Visual
3. Foreign matter	Visual (see 4.3.5.2)

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Minor

41. Burrs----------Visual

4.3.4.4 Cover (see Dwg.C7258821, covering a detail of Dwg.D7258818)

Categories and defects	Methods of inspection
Major	
1. Inside diameter at large end	Gage
2. Length from base to .300 datum diameter	Gage
3. Outside contour of tapered portion	Gage
4. Length from inside of nose to .300 datum diameter	Gage
5. Protective coating or cladding damaged or missing	Visual
6. Cracked, split, or deformed	Visual
7. Wall thickness, min.	Gage
8. Foreign matter	Visual (see 4.3.5.2)
Minor	
41. Wall thickness, max	Gage
42. Nose thickness	Gage
43. Burrs	Visual

4.3.4.5 Body (see Dwg. D7258820, covering a detail of Dwg.D7258818)

Categories and defects	Method of inspection
Critical	
a. Minimum groove profile	Gage
Major	
1. Major diameter of external thread	Gage
2. Pitch diameter of external thread	Gage
3. Pitch diameter of internal thread	Gage
4. Minor diameter of internal thread	Gage
5. Concentricity of booster holder cavity with pitch diameter of internal thread	Gage
6. Diameter of spring groove, min.	Gage
7. Length from nose to rear of cover crimping shoulder	Gage
8. Length from base to bottom of .37700 ball	Gage
9. Length from spring groove to rotor safety spring retainer shoulder	Gage
10. Finish of rotor seat	Measuring device
11. Protective coating damaged or missing	Visual
12. Shoulder diameter (after knurling)	Gage

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Major (cont)

13. Knurl missing or incomplete----------Visual
14. Profile of tapered surfaces from nose to crimping shoulder----------Gage
15. Chamfer at entrance to firing pin cavity-------Measuring device
16. Length of internal thread, min. (effective thread)----------Gage
17. Foreign matter in rotor seat or spring slot----Visual (see 4.3.5.2)
18. Burr ----------Visual
19. Spring slot width, min. ----------Gage
20. Length of spring slot, min.----------Gage
21. Location of spring slot from shoulder, min-----Gage
22. Concentricity of firing pin hole with pitch diameter of internal thread ----------Gage
23. Finish of firing pin hole----------Measuring device (see 4.3.5.2)
24. Diameter, cylindrical surface, nose end--------Gage

Minor

41. Length of thread relief at shoulder, max-------Gage
42. Length of thread relief at base ----------Gage
43. Total length ----------Gage
44. Finish, except firing pin hole ----------Visual
45. Radius at rear of knurled shoulder, max--------Gage
46. Depth of counterbore from base to start of internal thread----------Gage
47. Length of internal thread including counterbore and runout----------Gage
48. Diameter of booster holder cavity----------Gage
49. Centrality of spring slot with pitch diameter of external thread----------Gage
50. Wall thickness variation at nose ----------Gage

4.3.4.6 Body, spring, and spring retainer assembly (see Dwg. D7258819), covering a detail of Dwg. D7258818)

Categories and defects	Method of inspection
Critical	
a. Location of spring tip, more than .002 above .080 maximum dimension----------	Gage
b. Spring displacement ----------	Measuring device

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Major

1. Location of spring tip, .080 max ----------Gage
2. Location of spring tip, .090 min. (with minimum rotor in place)----------Gage
3. Loose or missing rotor safety spring retainer--Manual-Visual (see 4.3.5.1)
4. Gap between body shoulder and rotor safety spring retainer----------Gage
5. Diameter of firing pin hole----------Gage
6. Rotor safety spring retainer protruding above fuze body taper----------Gage
7. Burr ----------Visual
8. Crack in rotor safety spring retainer----------Visual
9. Foreign matter ----------Visual (see 4.3.5.2)

4.3.4.7 Firing pin (see C7258823, covering a detail of Dwg D7258818)

Categories and defects	Method of inspection
Critical	
a. Flange thickness, min	Gage
b. Flange missing	Visual
Major	
1. Diameter of flange	Gage
2. Large body diameter	Gage
3. Profile of point	Gage
4. Arming spring locking shoulder missing or improper	Visual
Minor	
41. Surface roughness	Visual
42. Body length from under side of flange, min.	Gage

4.3.4.8 Sealed packing container

Categories and defects	Method of inspection
Minor	
41. Container damaged	Visual
42. Marking incorrect, incomplete, illegible or missing	Visual

4.3.5 Defect standards

4.3.5.1 Any rotor safety spring retainer that moves when subjected to finger pressure (applied by twisting the retainer between the thumb and index finger), or that moves when a probe is applied against the side of the retainer at the spring slot will be classed defective.

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4.3.5.2 Any part or subassembly showing the presence of dirt, oil, grease, corrosion, or other foreign substance shall be classed defective.

4.3.6 Inspection equipment

4.3.6.1 Government verification inspection.-Government verification inspection will be accomplished with the use of gages and equipment that have been certified by the contracting officer, to the design specified in "Index to List of Inspection Gages and Parts Gaged on; Fuze, PD, M505A1, Metal Parts", GL7259308.

4.3.6.2 Contractor inspection.-Unless otherwise specified herein, contractor inspection will be accomplished with the use of gages and equipment designed to perform, to the required accuracy, the inspections and controls delineated in the contractor's written description specified in 4.1.1.1. In inspecting the contractor's inspection equipment the Government inspector will determine that the contractor has available and utilized correctly, gaging, measuring and test equipment of required accuracy and that the instruments are of proper type and range to make measurements of the required accuracy. The contractor shall have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration shall be maintained by the contractor and made available for review by the Government. The calibration of gages, standards, and instruments will be periodically checked by authorized Ordnance personnel.

4.4 Test methods.-The following tests shall be conducted by the contractor as specified in MIL-G-2550 in accordance with the methods specified below. Sampling and acceptance criteria shall be as specified with the test description.

4.4.1 Corrosion resistance.-The samples selected for the test shall be tested in accordance with Method 811 of Federal Test Method Standard No. 151.

4.4.2 Spring displacement.-The assembly shall be tested in the following manner: Place the assembly into Ordnance testing device, Dwg. D7259320, in manner prescribed and apply a load to the push ram to determine the resistance of the rotor safety spring to the push load. After the push ram test, each assembly shall be inspected for proper location of rotor safety spring tip. Assemblies may be tested without rotor safety spring retainers in place.

4.4.3 Spring embrittlement.-The rotor safety spring samples shall be tested by displacing the spring arm 120° from the spring base, using testing device shown on Dwg. D7259500. The sample springs tested shall not be returned to the lot.

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4.4.4 Hardness.- The sample fuze covers selected for this test shall be tested in accordance with Method 243 of Federal Test Method Standard No. 151, utilizing superficial Rockwell Hardness Testers and the 30T scale. The covers will be sectioned axially and the hardness impression will be taken on the inside surface at approximately the center of the specimen. When covers of alternate material are furnished, cladding shall be removed from the areas to be tested by a process which will not alter the hardness of the base metal.

4.5 Acceptance

4.51 At the beginning of the contract and as long as the Government inspector continues to perform the verification specified in 4.1.2.1, acceptance will be based on the results of the Government verification specified in 4.1.2.1 and 4.1.2.4.

4.5.2 After reduced Government verification specified in 4.1.2.2 has been initiated, acceptance will be based on the results of the contractor's inspection and quality control and the inspections reserved solely for performance by the Government and the results of the monitoring by the Government of the contractor's quality assurance program.

5. PREPARATION FOR DELIVERY

5.1 Packaging - Level C.-Packaging shall be in accordance with Specification MIL-P-10025.

5.2 Packing - Level C.-Packing shall be in accordance with Specification MIL-P-10025.

5.3 Marking.-Marking on packing containers shall be in accordance with Specification MIL-P-10025.

5.4 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Change in process.-The process includes all inspections, operations, screening, and sampling which are conducted by the contractor. A change in any contractor inspection or operation is construed as a change in process.

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6.3 AQL's.-The optional use of AQL values for either individual defects or classes of defects, as approved by the procuring activity prior to the initiation of production, is authorized to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the process.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian :

Army, Ordnance Corps

Preparing Activity:

Lake City Arsenal

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